SIMATS ENGINEERING
ASSESSMENT TOOL 2

COURSE NAME: Artificial Intelligence **COURSE CODE: CSA17**

COURSE OUTCOME COVERED

CO1: Apply AI basics such as intelligent agents and uninformed search methods to solve problems effectively.*(BL3)*

Assessment Tool Weightage: 35%

QUESTIONS: 3 **Duration : 1 Hour**
Total Marks:30

Annexure A

Constraint-Based Problem Solving

Code of Conduct:

I _____ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

Annexure A

Constraint-Based Problem Solving

1. A hospital needs to assign 3 doctors (D1, D2, D3) to 3 shifts (Morning, Afternoon, Night) such that:

   Constraints:

   i. D1 cannot work Night shift
   ii. D2 must work before D3 (Morning < Afternoon < Night)
   iii. Only one doctor per shift
   iv. D3 cannot work Morning
   v. All doctors must be assigned exactly one shift

   Apply Backtracking Search to find a valid assignment with all intermediate steps and constraint checks. State the final valid schedule

2. A robot operates in a grid environment (5×5). It must move from Start (S) to Goal (G). Obstacles block certain paths.

   Constraints:

   i. Movement allowed: Up, Down, Left, Right
   ii. Cost of each move = 1
   iii. Cannot pass through obstacles
   iv. Use Manhattan Distance heuristic

   Using above constraints and BFS Search Algorithm Show step-by-step node expansion and priority queue updates and determine the optimal path and cost**.**

3. An autonomous rescue robot is deployed in a disaster-affected building to locate and rescue survivors. The robot operates in a partially observable environment where some paths are blocked, and it must make decisions with limited information.

The building is represented as a grid of rooms. The robot starts at position S and must reach a survivor located at G.

Constraints:

- The robot can move up, down, left, right
- Some cells are blocked (obstacles) and cannot be traversed
- The robot has limited sensing capability (can only observe adjacent cells)
- Each movement has a cost of 1, but entering risky zones has an additional cost of 2
- The robot must minimize total cost while ensuring safe navigation
- The robot must update its knowledge dynamically (online search scenario)

Tasks:

a) Analyze all the given constraints and explain how they affect the robot's decision-making process.

b) Develop a solution approach using an appropriate search strategy (e.g., Uniform Cost Search / Online Search Agent). Clearly explain the steps involved.
c) Demonstrate how the robot applies AI concepts such as:

- Intelligent agents
- Rationality
- Environment type

d) Justify why your chosen approach is the most suitable for solving this problem under the given constraints.

RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Problem Understanding	25%	Clearly defines the problem and identifies all relevant constraints accurately	Identifies most constraints with minor gaps	Partial understanding; misses key constraints	Fails to identify constraints correctly
Constraint Analysis	25%	Analyzes constraints effectively and prioritizes them logically	Provides reasonable analysis with some prioritization	Limited analysis; weak prioritization	No meaningful analysis or prioritization
Solution Development	25%	Develops optimal and feasible solutions satisfying all constraints	Develops workable solutions with minor limitations	Solutions partially satisfy constraints	Solutions do not meet constraints
Application of Concepts	15%	Effectively applies appropriate concepts, methods, or tools	Applies concepts with minor errors	Limited application; requires guidance	Unable to apply relevant concepts
Justification	10%	Provides strong justification for decisions with logical reasoning	Provides justification with some clarity	Weak or unclear justification	No justification provided

ASSESSMENT TOOL 2 – ANSWERS

Constraint-Based Problem Solving

Answer 1 – Backtracking Search

Variables: D1, D2, D3. **Domain:** Morning (M), Afternoon (A), Night (N).

Constraints: D1 ≠ N; D2 must be before D3; one doctor per shift; D3 ≠ M; every doctor gets exactly one shift.

Backtracking steps:

Step	Assignment	Check	Result
1	D1 = M	D1 is not Night	Valid
2	D2 = A	D2 is before D3	Valid so far
3	D3 = N	D3 is not Morning; A < N	Valid

Final valid schedule: Morning – D1; Afternoon – D2; Night – D3.

All constraints are satisfied, so no further backtracking is required.

Answer 2 – Grid Search

The question specifies a 5×5 grid, four-direction movement, unit move cost, obstacle avoidance, and Manhattan distance. However, the uploaded source does not show the actual 5×5 grid containing the positions of S, G, and the obstacles. Therefore, an exact node-expansion sequence, priority-queue update sequence, optimal path, and numerical cost cannot be determined without inventing missing information.

Correct search procedure using the given constraints:

1. Put S in the frontier and mark its cost as 0.

2. Expand a node and generate only Up, Down, Left and Right neighbours.

3. Reject any neighbour that is outside the 5×5 grid or is an obstacle.

4. For each valid node, compute Manhattan distance $h(n) = |x – xG| + |y – yG|$.

5. With unit movement cost, update the frontier using the search priority. Continue expanding nodes until G is reached.

6. Reconstruct the path by following the parent of G back to S. The path cost equals the number of moves because each move costs 1.

Note: The source itself does not provide enough grid data to state a particular path or cost without making assumptions.

Answer 3(a) – Constraint Analysis

Movement constraint: The robot can move only up, down, left and right, so diagonal movement is not allowed.

Obstacle constraint: Blocked cells cannot be entered, so the robot must find a safe alternative route.

Limited sensing: The robot can observe only adjacent cells. It cannot know the complete environment in advance and must make decisions using current observations.

Cost constraint: Every normal movement costs 1, while entering a risky zone adds 2. Therefore, a shorter path is not always the cheapest path.

Safety and minimum-cost requirement: The robot must balance safe navigation with minimum total cost.

Dynamic knowledge: The robot updates its map as new cells are observed, making the problem an online-search scenario.

Answer 3(b) – Suitable Search Approach

Uniform Cost Search combined with an Online Search Agent is suitable.

1. Start at S with total cost 0.

2. Observe the adjacent cells and identify blocked and safe cells.

3. Add reachable cells to the frontier with their accumulated cost.

4. Select the frontier state having the lowest accumulated cost.

5. Move to that state and update the robot's knowledge with the newly observed neighbouring cells.

6. If a newly discovered obstacle blocks a planned route, update the frontier and choose another route.

7. Give risky cells their additional cost of 2, so entering such a cell has movement cost 1 + risk cost 2 = 3.

8. Continue until the robot reaches G. The resulting route has the minimum known total cost under the discovered constraints.

Answer 3(c) and 3(d) – AI Concepts and Justification

Intelligent agent: The rescue robot is an intelligent agent because it perceives the environment through its sensors, updates its knowledge, and selects actions to reach the survivor.

Rationality: A rational robot chooses actions that best achieve the goal while considering safety and total cost. It avoids blocked cells and prefers lower-cost safe routes.

Environment type: The environment is partially observable because the robot can observe only adjacent cells. It is also dynamic/uncertain from the robot's viewpoint because new information is discovered while moving. It is sequential because each action affects future decisions, and it is cost-based because movements have different total costs.

Justification: Uniform Cost Search is appropriate because movement costs are not necessarily identical once risky-zone costs are included. It expands the lowest-cost known path first. Combining it with online search allows the robot to update its knowledge as it discovers obstacles and safe/risky areas. This directly supports the requirements of minimum cost, safe navigation, limited sensing, and dynamic knowledge updates.

Final conclusion: The robot should use an online, cost-sensitive search strategy that continuously senses, updates its knowledge, and selects the safest minimum-cost route to G.